

Planning Page updated approach-

Now, we need to update the main architecture of the planning page pipeline. Major updates in regarding the building machine allocation, deciding of building primary SKU, building machine secondary SKU. And, round trip time calculation. No major changes in the curing side.

1. First i will analyse the last 10-12 days data of the plant, no which machine what are the dominant inch size- like 7001- 14, 13,15
7002- 15, 14, 12 like this.
2. After step1, we have to decide the SKUs on the building machine. Here comes our main logic. We will pick and start with the most dominant inch on a particular machine from the last 10-12 days data.
3. First we will filter out the same size eligible SKUs and which are running in the building side, on a building machine, then we will calculate it's consumption per shift, then we will check and calculate for how many hours this SKUs has already buffer available to it, means it will not go starvation for the next 4 hours, so till 4 hours assign the building machine to other SKUs which are going in the starvation, then i will calculate in how many hours this SKU can creates it's buffer so that it will not go to starvation for the next 8 hours when building machine leaves it. On this basis we will assign the SKUs building machines [now, their is no primary and no secondary SKUs on the building side]. Main aim is to reduce the starvation events. [SKUs which are not running- machine idle- should we need to produce them, i think no needs] machine will handle it auto when it goes to the running side.
4. Example for the explanation of the point 3- on a particular building machine- same size eligible SKUs which are also running-

SKUs Consumption Inventory present (hr. In terms curing) time for inventory buffer (8 hr.)

SKUs	Consumption (per shifts)	Inventory present (hr. In terms curing)	time for inventory buffer (8 hr.curing)
SKUa	400	4	4
SKUb	300	3	1
SKUc	200	6	4
SKUd	100	8	3

Now, how my building machine should select the SKUs, to provide the SKUs on time to curing press and to reduce starvation events as well.

Machine should picks the SKUb first

Timeline- starting 7AM

7 AM- 8AM- build SKUb [now, remaining 2 hours inventory + 8 hours more buffer maintained].

8 AM- 12 + (some time) [assume 12:30] PM- build SKUa- 8 hours buffer, 12 + some time because- 11-12 AM- old buffer consumed the curing taking from new buffer.

Now, we have 2 cases-

Case1-

12:30PM- (4:30PM + some extra time, let say) 5:PM- build it's buffer and extra time because curing started taking more the new buffer and to maintain it for the 8 hours needs to built it for the more time.

Case2-

12:30PM- 3:00PM build SKUc and then shifts to SKUd 3:00PM to 4:00 PM then shifted to SKUa or SKUd should be made till 3:00 PM to 6:00 PM

[These 2 cases tell us whether we wants to reduce the starvation so here we will have less building CO and whether we wants to eliminates the starvation cases but this will leads to too many building COs]. We have to decide this based on the analysis and testing.

Other important note points-

1. And, one very important note point- this 8 hours/1 shiit buffer is maintained and all calculations are performed on basis to this- so we have to decide this value as well. Whether is fixed for all the machine groups or variable as per machine group.
2. Now, their is no round trip calculation, primary and secondary SKUs approach in our pipeline, this is updated approach.
3. And, another important point, we can take 1 building different size CO on stage 1 (15 machines), BJ machine (6 machines), 0 on uni_narrow machines (3 machines), can take multiple on VMI machines [only when needed] (8 machines), and no limit on stage2 machines (built GTs from the carcass)) 6 machines- no limit of different size CO here, just maximise it's total utilization.
4. When the curing press goes to starvation, then it can produce less but it may have planned curing CO on the next day, but it should not happen, and in optimal way it should shift some shifts or days ahead, based on it's starvation events on the curing press.